DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

FUNCTIONAL REQUIREMENTS & VALIDATION

TI C2000 F28069M dual-inverter control, interfaces, evidence, and acceptance

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.1 | 5 AUGUST 2026

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Document scope	System behavior, operating modes, external interfaces, verification, and release boundary
Evidence basis	Editable Simulink models, handwritten ADC source, CAN interfaces, scripts, and retained evidence
Distribution	Customer-facing; controlled source models, custom firmware, and engineering evidence

DOCUMENT PURPOSE

Defines the controlled functional, performance, interface, safety, verification, and acceptance requirements for the public dual-motor dynamometer platform. Implementation detail and efficiency characterization are controlled in the companion engineering report.

1. Purpose, Scope, and Use of This FRD

This Functional Requirements Document (FRD) defines the required behavior, external interfaces, evidence obligations, constraints, and acceptance basis for the public Dual-Motor PMSM Dynamometer Platform. It is intended to support engineering review, controlled modification, verification planning, and customer understanding of the released system boundary.

Unlike the companion Embedded Control & Characterization Report, this document does not present characterization plots, processing methods, or measured-result interpretation. It establishes the required system behavior and the evidence needed to demonstrate compliance.

WHAT MAKES THIS AN FRD

Each mandatory capability is expressed as an atomic “shall” statement with a unique identifier, verification method, status, and retained evidence reference. Narrative sections explain system intent and operating behavior so the requirement tables are understandable without becoming an implementation manual.

1.1 Intended audience and decisions supported

- Customers and reviewers evaluating system capability and release maturity.
- Engineers modifying host, target, CAN, ADC, or acquisition functions.
- Test personnel selecting verification methods and acceptance evidence.
- Maintainers determining whether a proposed change affects an approved requirement or interface.

1.2 Public system boundary

The public release includes selected editable Simulink models, handwritten ADC peripheral source, MATLAB setup and analysis utilities, interface references, schematics, and approved evidence figures. Generated code, binaries, private raw data, vendor manufacturing packages, and institution-specific instructional materials remain outside the release.

2. System Overview and Operational Concept

The platform couples two encoder-equipped permanent-magnet synchronous machines on a common shaft. Motor 1 establishes shaft speed. Motor 2 applies opposing or assisting q-axis current so the coupled assembly can be exercised as a compact dynamometer. A TI C2000 F28069M target executes the real-time control loops, while supervisory models provide commands, acquisition, and test orchestration.

Figure 1. Public functional architecture and system boundary.

2.1 Functional decomposition

Element	Required responsibility	Boundary
Supervisory host	Provide speed, load/current, enable, selection, and test commands; collect observable signals.	Non-real-time supervision
C2000 target	Execute embedded FOC, encoder acquisition, CAN handling, ADC acquisition, and deterministic target logic.	Real-time control
Dual inverter stages	Convert DC bus power into controlled three-phase machine excitation.	TI reference hardware
Coupled PMSMs	Provide drive and load/assist behavior through a shared mechanical shaft.	Physical test article
Analysis workflow	Transform retained signals into reviewable engineering evidence.	Postprocessing

2.2 Normal operating sequence

A typical run begins with wiring and polarity verification, low-risk target deployment, encoder and phase-order confirmation, command enable, controlled speed and load application, data capture, and post-run evidence review. The host coordinates this sequence but does not replace the embedded timing or current-control functions.

3. Requirement Structure and Interpretation

Requirements are grouped by functional behavior, physical and communication interfaces, acquisition/evidence obligations, and release constraints. Every mandatory requirement uses “shall.” Explanatory paragraphs provide intent and context; they are not substitutes for the numbered requirement statements.

Attribute	Meaning
Identifier	Stable reference used in design discussions, verification records, and change review.
Requirement	Atomic mandatory behavior written as a testable “shall” statement.
Rationale	Why the capability is necessary to the system mission or evidence chain.
Verification	Inspection, analysis, demonstration, test, or a controlled combination.
Status	Demonstrated, Verified, Open, or Not Claimed for this public release.
Evidence	Retained model, source file, figure, procedure, or repository artifact supporting closure.

STATUS CONVENTION
“Demonstrated” means operation has been shown on the engineering platform. “Verified” means release content or an analysis obligation has been checked. Neither term implies calibrated metrology, production qualification, functional-safety certification, or regulatory approval.

3.1 Operating modes

Mode	Description	Entry/exit expectation
Disabled / setup	Commands are configured while torque-producing operation remains disabled.	Default safe software state; wiring and feedback checks occur here.
Drive / load operation	Motor 1 regulates shaft speed while Motor 2 applies a commanded load or assist current.	Entered only after enable and feedback checks; exited by disable or fault response.
CAN-supervised	Command words arrive over the documented CAN path and	Communication must

Mode	Description	Entry/exit expectation
operation	are returned or observed diagnostically.	be validated before relying on remote commands.
Acquisition / review	Signals are captured, exported, and processed after or during a controlled run.	Evidence must remain traceable to source models and scripts.

4. Functional Requirements

The functional requirements define the externally observable behavior of the platform. Implementation details may change, but the required drive/load roles, command paths, feedback separation, and evidence workflow must remain intact unless the FRD is revised.

ID	Area	Requirement	Rationale	Verification	Status
SYS-001	Real-time control	The system shall execute fast motor-control loops on the embedded target independently of host display timing.	Host timing is unsuitable as the deterministic control clock.	Inspection + functional test	Demonstrated
SYS-002	Dual-machine operation	The system shall command one PMSM as the drive machine and the second as a controllable load or assist machine.	Defines the dynamometer function.	Functional test	Demonstrated
SYS-003	Speed reference	The system shall accept a supervisory speed reference for the drive machine.	Provides controlled operating-point selection.	Functional test	Demonstrated
SYS-004	Load reference	The system shall accept a q-axis current or equivalent load reference for the load machine.	Provides controllable shaft loading.	Functional test	Demonstrated
SYS-005	Enable control	The system shall provide explicit enable and disable control for commanded operation.	Separates configuration from torque-producing operation.	Inspection + test	Demonstrated
SYS-006	Feedback	The system shall maintain independent encoder feedback paths for both machines.	Prevents cross-coupled or ambiguous machine state.	Inspection + feedback test	Demonstrated
SYS-007	Observability	The system shall expose speed, current, voltage, torque-related, power-related, and diagnostic quantities required by the approved workflow.	Supports verification and engineering review.	Inspection + analysis	Demonstrated
SYS-008	Repeatable workflow	The host workflow shall support repeatable setup, capture, export, and postprocessing.	Reduces ad hoc operator dependence.	Procedure review	Demonstrated

5. Interface Requirements

Interface requirements control the points where errors are most likely to damage hardware, invalidate data, or break compatibility. They cover power, phase and encoder routing, CAN communication, and the custom analog acquisition path.

Figure 2. BoosterPack-to-LaunchPad board-interface reference retained with the release.

ID	Interface	Requirement	Verification	Status
IF-001	DC input	The inverter stack shall use a verified 24 VDC supply with current limiting during bring-up.	DMM + startup test	Demonstrated
IF-002	Motor phases	Each PMSM shall be connected consistently to its assigned inverter phase outputs.	Continuity + spin test	Demonstrated
IF-003	Encoder routing	Motor 1 and Motor 2 encoder channels shall remain electrically and logically distinct.	Pin inspection + feedback test	Demonstrated
IF-004	CAN physical layer	The CAN bus shall provide CANH, CANL, reference ground, and appropriate termination.	Resistance + bus test	Demonstrated
IF-005	CAN command	The target shall receive the four-word supervisory command payload on the documented command identifier.	Bus test	Demonstrated
IF-006	CAN diagnostics	The target shall provide a return identifier for command echo or telemetry diagnostics.	Bus test	Demonstrated
IF-007	Custom ADC acquisition	The target shall configure two additional simultaneous ADC input pairs using handwritten F28069M peripheral source.	Source inspection + acquisition test	Demonstrated

INTERFACE CONTROL NOTE
Exact header positions, polarity, termination, and board-revision pin assignments must be confirmed against the current retained schematics and interface summary before energizing hardware.

6. Data, Evidence, and Verification Requirements

This FRD does not reproduce characterization figures or analyze measured performance. It requires every published engineering claim to remain traceable to retained source artifacts and sufficient verification evidence.

ID	Area	Requirement	Verification	Status
DAT-001	Evidence	Published engineering results shall be traceable to retained	Repository	Verified

ID	Area	Requirement	Verification	Status
	traceability	scripts, source models, or approved evidence outputs.	inspection	
DAT-002	Source distinction	The release shall distinguish handwritten firmware, editable Simulink source, generated support code, and third-party reference material.	Repository + document inspection	Verified
DAT-003	Result ownership	Detailed characterization methods, plots, and measured-result interpretation shall reside only in the companion implementation report.	Document inspection	Verified
DAT-004	Release integrity	The release shall include checksums and shall open directly at repository root without an unnecessary outer directory.	Archive inspection	Verified

6.1 Verification strategy

Verification answers whether the implementation satisfies a stated requirement. Inspection is used for model structure, source configuration, repository boundaries, and wiring documentation. Functional and bus tests are used where observable behavior must be exercised. Analysis is used only where equations, signal derivations, or retained outputs must be evaluated.

Method	Applied to	Acceptance principle
Inspection	Models, source, schematics, repository structure	Required element exists, is correctly configured, and matches the controlled interface.
Demonstration	Operator workflow and observable behavior	Capability can be shown under controlled conditions without exhaustive measurement.
Functional test	Drive/load operation, enable, feedback, acquisition	Commanded input produces the expected observable system behavior.
Bus test	CAN receive, unpacking, return/diagnostic traffic	Frames are received and interpreted on the documented identifiers and payload layout.
Analysis	Signal availability and retained evidence chain	Calculation or review supports the stated requirement without unsupported extrapolation.

7. Requirements Traceability Matrix

The traceability matrix connects each requirement to its verification method and retained evidence. It is the primary closure record for this public FRD. Detailed implementation discussion remains in the companion report.

Requirement	Verification method	Retained evidence	Status
SYS-001	Inspection + functional test	C2000_Dual_Motor_CAN_Target.slx	Demonstrated

Requirement	Verification method	Retained evidence	Status
SYS-002	Functional test	target and host models	Demonstrated
SYS-003	Functional test	host setup models	Demonstrated
SYS-004	Functional test	host setup models	Demonstrated
SYS-005	Inspection + test	host/target models	Demonstrated
SYS-006	Inspection + feedback test	board interface + target model	Demonstrated
SYS-007	Inspection + analysis	target model + processing scripts	Demonstrated
SYS-008	Procedure review	setup/logging/analysis scripts	Demonstrated
IF-001	DMM + startup test	INTERFACE_SUMMARY.md	Demonstrated
IF-002	Continuity + spin test	INTERFACE_SUMMARY.md	Demonstrated
IF-003	Pin inspection + feedback test	board-interface figure	Demonstrated
IF-004	Resistance + bus test	CAN model set	Demonstrated
IF-005	Bus test	CAN_Receive.slx	Demonstrated
IF-006	Bus test	CAN_Transmit.slx	Demonstrated
IF-007	Source inspection + acquisition test	F28069M_ADC_Custom_Config.c	Demonstrated
DAT-001	Repository inspection	models, scripts, reports, checksums	Verified
DAT-002	Repository + document inspection	embedded-control/README.md + NOTICE.md	Verified
DAT-003	Document inspection	FRD + companion report	Verified
DAT-004	Archive inspection	SHA256SUMS.txt + final ZIP	Verified

8. Constraints, Assumptions, and Safety Requirements

8.1 Assumptions and dependencies

- Compatible MATLAB/Simulink, C2000 support packages, and required hardware drivers are available to the user.
- The retained TI schematics and connector references correspond to the board revisions being operated.
- Mechanical coupling, motor mounting, and encoder attachment are secure before commanded rotation.
- Instrumentation and calculated quantities are engineering evidence unless separately calibrated and certified.

8.2 Safety and release constraints

ID	Constraint	Acceptance statement
SAF-001	Polarity	DC polarity shall be verified before either inverter stage is energized.
SAF-002	Conservative startup	Initial operation shall use low speed, zero load command, conservative current limits, and accessible shutdown.
SAF-003	Feedback validation	Encoder direction and phase order shall be confirmed before closed-loop sweeps.
SAF-004	Independent protection	Software disable shall not be represented as an independent safety function.
REL-001	Qualification boundary	The release shall not be represented as production-qualified, calibrated, EMC-qualified, or safety-certified.
REL-002	IP boundary	Generated binaries, private raw data, vendor manufacturing packages, and institution-specific instructional material shall remain excluded.

9. Validation Summary and Release Gate

Validation asks whether the released platform is useful for its intended engineering purpose. The public evidence supports coupled drive/load operation, supervisory CAN command flow, additional ADC acquisition, repeatable host workflows, and traceable characterization. It does not establish product certification or calibrated laboratory metrology.

Validation objective	Acceptance basis	Status
Coupled drive/load operation	Motor 1 establishes shaft speed while Motor 2 applies controlled loading.	Demonstrated
CAN supervisory control	Command receipt and diagnostic return are represented in retained models and evidence.	Demonstrated
Custom ADC path	Additional PWM-triggered simultaneous input pairs are configured and integrated.	Demonstrated
Evidence traceability	Published claims map to retained models, custom source, scripts,	Verified

Validation objective	Acceptance basis	Status
	figures, and checksums.	
Production qualification	Calibration, endurance, EMC, environmental, cybersecurity, and independent safety validation.	Not claimed

RELEASE GATE
Publish only after both reports are rendered page by page, figure captions and tables are visually checked, links resolve, checksums are regenerated, the ZIP opens at repository root, and no private or institution-specific instructional material remains.